

類別資料分析 第2章

The Analysis of Categorical Data Chapter 2 : Two-way Contingency Tables

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Two-way Contingency Tables

*列連表 (Contingency Table) : 兩個類別變數，X 和 Y，X 有 I 種可能結果，Y 有 J 種可能結果。

結合 X 和 Y，就有 $I \times J$ 種可能，稱作 $I \times J$ 列連表。

結合 2 個變數的列連表，稱作 2 因子列連表 (Two-way Table)。

結合 3 個變數的列連表，稱作 3 因子列連表 (Three-way Table)。

e. g.

		Y : 是否相信有來生		
		Yes (Y=1)	No (Y=2)	Total
X : 性 別	Females (X=1)	435	147	582
	Males (X=2)	375	134	509
	Total	810	281	1091

$$X = \begin{cases} 1 & \text{Females} \\ 2 & \text{Males} \end{cases}$$

$$Y = \begin{cases} 1 & \text{Yes} \\ 2 & \text{No} \end{cases}$$

Two-way Contingency Tables

* $P_{ij} = P(X = i, Y = j)$: the joint distribution of X and Y

$\sum_{i=1}^2 \sum_{j=1}^2 P_{ij} = 1$ P_{i+} & P_{+j} : the marginal probability

$P_{1+} = P_{11} + P_{12}, P_{2+} = P_{21} + P_{22},$ i. e. $P_{i+} = P(X = i) = \sum_{j=1}^2 P_{ij}$

$P_{+1} = P_{11} + P_{21}, P_{+2} = P_{12} + P_{22},$ i. e. $P_{+j} = P(Y = j) = \sum_{i=1}^2 P_{ij}$

Y : the column variable, response variable

X : the row variable, explanatory variable

		Y		Total
		Y=1	Y=2	
X	X=1	P_{11}	P_{12}	P_{1+}
	X=2	P_{21}	P_{22}	P_{2+}
Total		P_{+1}	P_{+2}	1

Two-way Contingency Tables

* n_{ij} : the cell count

$\sum_{i=1}^2 \sum_{j=1}^2 n_{ij} = n$: the total sample size

$\hat{P}_{ij} = \frac{n_{ij}}{n}$: the sample proportion

Y : the column variable, response variable

X : the row variable, explanatory variable

		Y		Total
		Y=1	Y=2	
X	X=1	n_{11}	n_{12}	n_{1+}
	X=2	n_{21}	n_{22}	n_{2+}
Total		n_{+1}	n_{+2}	n

Two-way Contingency Tables

e. g.

		Y : 是否相信有來生		
		Yes (Y=1)	No (Y=2)	
X : 性 別	Females (X=1)	$\hat{P}_{11} = 435/1091$	$\hat{P}_{12} = 147/1091$	$\hat{P}_{1+} = 582/1091$
	Males (X=2)	$\hat{P}_{21} = 375/1091$	$\hat{P}_{22} = 134/1091$	$\hat{P}_{2+} = 509/1091$
		$\hat{P}_{+1} = 810/1091$	$\hat{P}_{+2} = 281/1091$	

the conditional probability of Y, given X : $\hat{P}(Y = 1|X = 1) = \frac{435}{582} \approx 0.747$

$$\hat{P}(Y = 2|X = 1) = \frac{147}{582} \approx 0.253$$

$$\hat{P}(Y = 1|X = 2) = \frac{375}{509} \approx 0.737$$

$$\hat{P}(Y = 2|X = 2) = \frac{134}{509} \approx 0.263$$

Two-way Contingency Tables

*Independent : Two random variables X and Y are independent if

$$P(X = i, Y = j) = P(X = i) \cdot P(Y = j) \text{ or } P(Y = j|X = i) = P(Y = j)$$

$$P(Y = j|X = i) = \frac{P(X=i, Y=j)}{P(X=i)}$$

i. e. $H_0: P_{ij} = P_{i+} \cdot P_{+j}$

e. g. $P(\text{Yes}|\text{Females}) = P(\text{Yes}|\text{Males}) \Leftrightarrow \begin{cases} P(Y = 1|X = 1) = P(Y = 1) = P_1 \\ P(Y = 1|X = 2) = P(Y = 1) = P_1 \end{cases}$

若男生相信有來世的比例和女生相信有來世的比例相同。

則是否相信有來世 (Y) 和 性別 (X) 為獨立的兩個變項。

Two-way Contingency Tables

*Poisson Sampling (Fixed time)

$n_{ij} \sim \text{Poi}(\lambda_{ij})$, 4個 Random variables, 4 個 Parameters

e. g. H_0 : 是否飲酒和交通事故是否造成死亡無關

$$\equiv H_0 : OR = 1$$

*Binomial Sampling (Given n_{1+} , n_{2+})

$n_{11} * n_{1+} \sim \text{Bin}(n_{1+}, P_1), P_1 = P(Y = 1 * X = 1)$

$n_{21} * n_{2+} \sim \text{Bin}(n_{2+}, P_2), P_2 = P(Y = 1 * X = 2)$

e. g. H_0 : 服用維他命與否和是否感冒與否無關

$$\equiv H_0 : P_1 = P_2$$

		交通事故是否造成死亡		
		死亡	非死亡	
是否飲酒	飲酒	n_{11}	n_{12}	n_{1+}
	未飲酒	n_{21}	n_{22}	n_{2+}
		n_{+1}	n_{+2}	

		Y		
		感冒	沒感冒	
X	維他命	n_{11}	n_{12}	n_{1+}
	安慰劑 (Placebo)	n_{21}	n_{22}	n_{2+}

Two-way Contingency Tables

*Multinomial Sampling (Fixed $n_{++} = n$)

$(n_{11}, n_{12}, n_{21}, n_{22}) \sim \text{Mul}(n, P_{11}, P_{12}, P_{21}, P_{22})$

抽樣條件：(1) 不放回隨機抽樣

(2) 每個人被抽到的機率相等

e. g. H_0 : 性別和是否相信有來生無關

		是否相信有來生		
		是	否	
性別	女生	n_{11}	n_{12}	n_{1+}
	男生	n_{21}	n_{22}	n_{2+}
		n_{+1}	n_{+2}	n_{++}

Two-way Contingency Tables

*Test for the difference of proportions

$$\text{Let } \hat{P}_1 = \frac{n_{11}}{n_{1+}}, \hat{P}_2 = \frac{n_{21}}{n_{2+}}$$

(1) $H_0: P_1 = P_2$ vs. $H_1: P_1 \neq P_2$ (two-tailed)

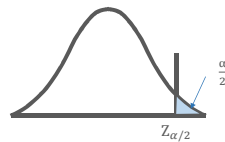
$$(2) Z = \frac{\hat{P}_1 - \hat{P}_2 - (P_1 - P_2)}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_{1+}} + \frac{1}{n_{2+}}\right)}} \sim N(0, 1) \text{ under } H_0: P_1 = P_2 = P, \text{ where } \hat{P} = \frac{n_{11} + n_{22}}{n_{1+} + n_{2+}}$$

(3) Reject H_0 if $Z > 1.96$ or $Z < -1.96$ (significance level $\alpha = 0.05$)

Two-way Contingency Tables

*A large sample $100(1-\alpha)\%$ confidence interval for $P_1 - P_2$:

$$\hat{P}_1 - \hat{P}_2 \pm Z_{\left(\frac{\alpha}{2}\right)} \hat{\sigma}(\hat{P}_1 - \hat{P}_2), \text{ where } \hat{\sigma}(\hat{P}_1 - \hat{P}_2) = \sqrt{\frac{\hat{P}_1(1-\hat{P}_1)}{n_{1+}} + \frac{\hat{P}_2(1-\hat{P}_2)}{n_{2+}}}$$



Two-way Contingency Tables

e. g. 雙盲實驗 (Double blind) : 在研究中的醫生和病患都不知道服用的是哪一種藥

(1) H_0 : 服用阿斯匹靈是否心肌梗塞無關 $\Leftrightarrow H_0: P_1 = P_2$

$P_1 = P(\text{心肌梗塞} * \text{安慰劑}), P_2 = P(\text{心肌梗塞} * \text{阿斯匹靈})$

(2) $\hat{P}_1 = \frac{189}{11034} \approx 0.0171, \hat{P}_2 = \frac{104}{11037} \approx 0.0094$

$$\hat{\sigma}(\hat{P}_1 - \hat{P}_2) = \sqrt{\frac{0.0171(1-0.0171)}{11034} + \frac{0.0094(1-0.0094)}{11037}} \approx 0.0015$$

		心肌梗塞		
		Yes	No	
服用藥物	安慰劑	189	10845	11034
	阿斯匹靈	104	10933	11037

(3) A 95% confidence interval for $P_1 - P_2$ is $(0.0171 - 0.0094) \pm 1.96 \cdot 0.0015 \approx (0.005, 0.011)$

$\because 0 \notin (0.005, 0.011) \therefore \text{Reject } H_0 \Rightarrow \text{故服用阿斯匹靈可降低心肌梗塞發生的機率}$

*The relative risk = $\frac{\hat{P}_1}{\hat{P}_2} = \frac{0.0171}{0.0094} \approx 1.82 \Rightarrow \text{A 95\% C. I. for } \frac{P_1}{P_2} \text{ is } (1.43, 2.30)$

Two-way Contingency Tables

*Types of Observational Studies

(1) 前瞻性研究 (或稱世代研究) Prospective study (or Cohort study):

研究過程是將某一範圍的健康人分成兩組，一組暴露於某種因素，另一組未暴露，

然後一段時間後觀察比較兩組人生病或死亡的情形，以判定某種因素與疾病發生的關係。

· 注意：此研究假設每個人患病的機會一樣，在研究初，應盡可能使暴露組與非暴露組其他的

背景因素 (如：年齡、性別等) 一致，才不至於干擾研究結果。

e. g. 服用阿斯匹靈發生心肌梗塞的機率 和 服用安慰劑發生心肌梗塞的機率 是否相同

Two-way Contingency Tables

*Types of Observational Studies

(2) 回溯性研究 (或病例—對照研究) Retrospective study (or Case - control study) :

此研究先選定病例組，並設相對應的對照組，在兩組人中回溯調查以往是否暴露於某種因素，進行兩組的比較分析，以推測這種因素是否與這種疾病相關。

e. g. 抽菸和肺癌是否有關

(3) 橫斷面研究 Cross – sectional study

抽樣和分類同時進行

- 以上三類稱作 **Observational studies**，通常來自生物醫學或社會科學所收集的觀測資料，而 **clinical trial** 為 **experimental study**，研究者可控制實驗誤差。

Two-way Contingency Tables

*Relative Risk : the ratio of the “success” probabilities for the two groups in 2×2 tables

$$\text{i. e. the relative risk} = \frac{P_1}{P_2} \Rightarrow \text{the sample relative risk} = \frac{\hat{P}_1}{\hat{P}_2}$$

- 當 P_1 和 P_2 都很接近 0 時，用比例差可能造成誤導，應採用 **Relative risk**。

- 若一組資料比例為 $\hat{P}_1 = 0.01, \hat{P}_2 = 0.001$ ，另一組資料比例為 $\hat{P}_1 = 0.41, \hat{P}_2 = 0.401$ ，兩組資料比例差皆為 0.009，但顯然有一資料值得注意的，

因為吃 A 藥有副作用的比例是吃 B 藥的 10 倍，此時用 **Relative risk** 來比較較好。

$$\text{e. g. 第一組資料的 relative risk} = \frac{0.1}{0.001} = 10, \text{第二組資料的 relative risk} = \frac{0.41}{0.401} \approx 1.02$$

- 若 $P_1 = P_2$ ，則 **relative risk** = 1，即解釋變數和反應變數獨立。

		Y : 副作用	
		Yes	No
X	A 藥	P_1	$1-P_1$
	B 藥	P_2	$1-P_2$

Two-way Contingency Tables

*Odds : the odds of the “success” are defined as :

$$\text{odds 1} = \frac{P_1}{1-P_1}; \text{odds 2} = \frac{P_2}{1-P_2},$$

		Y		
		Yes	No	
X	1	P_1	$1-P_1$	1
	2	P_2	$1-P_2$	1

$$\hat{P}_1 = \frac{n_{11}}{n_{1+}} \Rightarrow \widehat{\text{odds 1}} = \frac{\hat{P}_1}{1-\hat{P}_1} = \frac{n_{11}}{n_{12}}$$

$$\hat{P}_2 = \frac{n_{21}}{n_{2+}} \quad \widehat{\text{odds 2}} = \frac{\hat{P}_2}{1-\hat{P}_2} = \frac{n_{21}}{n_{22}}$$

where P_1 and P_2 are the probabilities of success in row 1 and 2, respectively.

$$\cdot \text{odds} = \frac{P}{1-P} \Rightarrow P = \frac{\text{odds}}{\text{odds}+1}$$

當 $\text{odds 1} = \text{odds 2}$ ($P_1 = P_2$)，代表 X 和 Y 兩變數獨立。

當 $\text{odds} = 4.0$ ，則 $P = \frac{4}{4+1}$ ，即成功機率 0.8，失敗機率 0.2。

i. e. 每觀測 1 個失敗，期望會觀測到 4 個成功。

當 $\text{odds} = \frac{1}{4}$ ，則 $P = \frac{1/4}{1/4+1}$ ，即成功機率 0.2，失敗機率 0.8。

i. e. 每觀測 4 個失敗，期望會觀測到 1 個成功。

Two-way Contingency Tables

$$\text{*Odds Ratio : the odds ratio} = \theta = \frac{\text{odds 1}}{\text{odds 2}} = \frac{P_1/(1-P_1)}{P_2/(1-P_2)} = \frac{P_{11}P_{22}}{P_{12}P_{21}}$$

· properties of the odds ratio :

(1) 當 $\theta = 1$ 代表 X 和 Y 兩變數獨立 (i. e. $P_1 = P_2$)

$\theta > 1$ 代表 X 和 Y 兩變數正相關 (i. e. $P_1 > P_2$)

$\theta < 1$ 代表 X 和 Y 兩變數負相關 (i. e. $P_1 < P_2$)

$$P_1 = P(\text{Success}|X=1) = \frac{P_{11}}{P_{1+}}$$

$$P_2 = P(\text{Success}|X=2) = \frac{P_{21}}{P_{2+}}$$

(2) 當行或列其中之一的類別順序交換時，會得到相同的 $|\log \theta|$ ，只是符號改變。
(Invariant property)

$$\theta = \frac{P_{11}P_{22}}{P_{12}P_{21}} \Rightarrow \theta' = \frac{P_{12}P_{21}}{P_{11}P_{22}} = \frac{1}{\theta}$$

$$\text{e. g. } \theta = 0.25, \theta' = \frac{1}{0.25} = 4$$

		Y	
		1	2
X	1	P_{11}	P_{12}
	2	P_{21}	P_{22}

 $\longrightarrow$

		Y	
		2	1
X	1	P_{12}	P_{11}
	2	P_{22}	P_{21}

Two-way Contingency Tables

*properties of the odds ratio :

(3) 將行或列任乘一數，不改變 odds ratio 之值。(i. e. independent of sample size)

(4) 當行變數或列變數交換，不改變 odds ratio之值。

*the odds ratio = $\frac{P_{11}/P_{12}}{P_{21}/P_{22}} = \frac{P_{11}P_{22}}{P_{12}P_{21}}$ (又稱作 the cross-product ratio)

the sample odds ratio $\hat{\theta} = \frac{n_{11}/n_{12}}{n_{21}/n_{22}} = \frac{n_{11}n_{22}}{n_{12}n_{21}}$

* $\hat{\theta}$ is the MLE of θ for the standard sampling schemes.

		Y	
		1	2
X	1	rP_{11}	rP_{12}
	2	P_{21}	P_{22}

		X	
		1	2
Y	1	P_{11}	P_{21}
	2	P_{12}	P_{22}

		Y	
		1	2
X	1	n_{11}	n_{12}
	2	n_{21}	n_{22}

Two-way Contingency Tables

e. g. odds 1 = $\frac{n_{11}}{n_{12}} = \frac{189}{10845} \approx 0.0174$

服用安慰劑的實驗者，平均每觀測 100 個沒有心肌梗塞者，
期望會觀測到 1.74 個有心肌梗塞者。

odds 2 = $\frac{n_{21}}{n_{22}} = \frac{104}{10933} \approx 0.0095$

服用阿斯匹靈的實驗者，平均每觀測 100 個沒有心肌梗塞者，
期望會觀測到 0.95 個有心肌梗塞者。

$\hat{\theta} = \frac{0.0174}{0.0095} = \frac{189 \times 10933}{104 \times 10845} \approx 1.832$

		心肌梗塞		
		Yes	No	
服用藥物	安慰劑	189	10845	11034
	阿斯匹靈	104	10933	11037

Two-way Contingency Tables

*For small to moderate sample sizes, the sampling distribution of the odds ratio is highly skewed.

The log odds ratio is symmetric about 0.

* $\log \hat{\theta} \sim N(\log \theta, \text{Var}(\log \hat{\theta}))$ or $\frac{\log \hat{\theta} - \log \theta}{\sqrt{\widehat{\text{Var}}(\log \hat{\theta})}} \rightarrow N(0, 1)$ (By Taylor's expansion)

$$\text{Var}(\log \hat{\theta}) \approx \left(\frac{1}{p_{11}} + \frac{1}{p_{12}} + \frac{1}{p_{21}} + \frac{1}{p_{22}} \right) \cdot \frac{1}{n} \Rightarrow \widehat{\text{Var}}(\log \hat{\theta}) = \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}} (\because \hat{p}_{ij} = \frac{n_{ij}}{n})$$

Two-way Contingency Tables

*A large-sample approximate confidence interval for $\log \theta$ is :

$$\log \hat{\theta} \pm Z_{\frac{\alpha}{2}} \cdot \text{ASE}(\log \hat{\theta}), \text{ where } \text{ASE}(\log \hat{\theta}) = \sqrt{\widehat{\text{Var}}(\log \hat{\theta})} = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$$

$\Rightarrow$ A large-sample confidence interval for θ is :

ASE : Asymptotic Standard Error

$$\left(\exp\left(\log \hat{\theta} - Z_{\frac{\alpha}{2}} \cdot \text{ASE}(\log \hat{\theta})\right), \exp\left(\log \hat{\theta} + Z_{\frac{\alpha}{2}} \cdot \text{ASE}(\log \hat{\theta})\right) \right)$$

Two-way Contingency Tables

e. g. 心肌梗塞案例

$$\log \hat{\theta} = \log(1.832) \approx 0.605$$

$$ASE(\log \hat{\theta}) = \sqrt{\left(\frac{1}{189} + \frac{1}{10933} + \frac{1}{10845} + \frac{1}{104}\right)} \approx 0.123$$

$$\Rightarrow \text{A 95\% confidence interval for } \log \hat{\theta} \text{ is } (0.605 \pm 1.96 \cdot 0.123) \approx (0.365, 0.846)$$

$$\text{A 95\% confidence interval for } \hat{\theta} \text{ is } (\exp(0.365), \exp(0.846)) \approx (1.44, 2.33)$$

$$\because 0 \notin (0.365, 0.846) \Leftrightarrow 1 \notin (1.44, 2.33) \quad \therefore \text{Reject } H_0 : \theta = 1 \Leftrightarrow H_0 : \log \theta = 0$$

i. e. 服用安慰劑發生心肌梗塞的勝算較服用阿斯匹靈發生心肌梗塞的勝算至少高出 44%

Note : 1.832 並非 (1.44, 2.33) 的中點，因為 $\hat{\theta}$ 的分配非對稱。

Two-way Contingency Tables

*The sample odds ratio $\hat{\theta} = 0$ or ∞ if $n_{ij} = 0$.

$$\text{The amended sample odds ratio } \tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$$

e. g. 心肌梗塞案例

$$\tilde{\theta} = \frac{189.5 \times 10933.5}{10845.5 \times 104.5} \approx 1.828 \approx \hat{\theta} = 1.832$$

Note : 1.832 並非 (1.44, 2.33) 的中點，因為 $\hat{\theta}$ 的分配非對稱。

Two-way Contingency Tables

* $\hat{\theta} = 1.832$ 不代表 $P_1 = 1.832 \cdot P_2$ ($\because$ Relative Risk = $\frac{P_1}{P_2}$)

$$\hat{\theta} = \frac{P_1/(1-P_1)}{P_2/(1-P_2)} = \text{Relative Risk} \times \frac{1-P_2}{1-P_1}$$

When the proportion of successes is close to 0 (i. e. $P_1, P_2 \approx 0$)

$$\Rightarrow \hat{\theta} \approx \frac{P_1}{P_2} = \text{Relative Risk} (\because 1 - P_1, 1 - P_2 \approx 1)$$

e. g. 心肌梗塞案例

$$\text{Relative Risk} = \frac{P_1}{P_2} = \frac{0.0171}{0.0094} \approx 1.82 \approx \hat{\theta} = 1.83$$

Two-way Contingency Tables

e. g. 心肌梗塞與抽菸是否有關：

$P(\text{Case}|\text{Yes})$
 $P(\text{Case}|\text{No})$ } 不可估計

由 1983-1988 年北義大利 69 歲以下罹患急性心肌梗塞的婦女 262 人作為病例組，每一病例

在相同醫院選 2 個其他病的病人配對作為控制組。

$H_0 : P(\text{Case}|\text{Yes}) = P(\text{Case}|\text{No})$ 無法檢定 ($\because \frac{262}{262+519} \approx \frac{1}{3}$ 沒有意義)

$\Rightarrow H_0 : P(\text{Yes}|\text{Case}) = P(\text{Yes}|\text{Controls}) \equiv H_0 : \theta = 1$ 可以檢定

$\because$ fixed n_{+1}, n_{+2}

		Y: 心肌梗塞	
		Case	Controls
X 曾抽菸否	Yes	172	173
	No	90	346
		262	519

Two-way Contingency Tables

$$*n_{11} \sim \text{Bin}(n_{+1}, P_1), n_{12} \sim \text{Bin}(n_{+2}, P_2),$$

where $P_1 = P(\text{Yes}|\text{Case})$ and $P_2 = P(\text{Yes}|\text{Controls})$

$$\Rightarrow \hat{P}_1 = \frac{n_{11}}{n_{+1}} = \frac{172}{262} = 0.656, \hat{P}_2 = \frac{n_{12}}{n_{+2}} = \frac{173}{519} = 0.333$$

		Y	
		Case	Controls
X	Yes	n_{11}	n_{12}
	No	n_{21}	n_{22}
		n_{+1}	n_{+2}

*Since n_{11} and n_{12} are dependent,

we can not calculate the test statistic $Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_{+1}} + \frac{1}{n_{+2}}\right)}} \sim N(0, 1)$ under $H_0 : P_1 = P_2 = P$,

$$\text{where } \hat{P} = \frac{n_{11} + n_{12}}{n_{+1} + n_{+2}}$$

Two-way Contingency Tables

$$*\hat{\theta} = \frac{0.656/(1-0.656)}{0.333/(1-0.333)} = \frac{172 \times 346}{173 \times 90} \approx 3.82$$

$$\Rightarrow \text{odds 1} = \frac{0.656}{1-0.656} \text{ 是 } \text{odds 2} = \frac{0.333}{1-0.333} \text{ 的 } 3.82 \text{ 倍}$$

$$\text{若 } P(\text{Case}|X) \approx 0 \cdot \hat{\theta} \approx \text{Relative Risk} = \hat{P}(\text{Case}|\text{Yes})/\hat{P}(\text{Case}|\text{No})$$

則我們可說抽菸發生心肌梗塞的機率是不抽菸發生心肌梗塞機率的 3.82 倍

Note : 原本的 Relative risk 不可估計

Two-way Contingency Tables

*The Pearson Chi-square statistic :

(1) $H_0 : P_{ij} = P_{i+} \cdot P_{+j} \equiv H_0 : X \text{ is independent of } Y$

(2) $X^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}} \sim \chi^2_{(I-1)(J-1)}$ under H_0 if $\hat{\mu}_{ij} \geq 5$,

where $\hat{\mu}_{ij} = \hat{E}(n_{ij}) = n \cdot \hat{P}_{ij} = n \cdot \hat{P}_{i+} \cdot \hat{P}_{+j} = n \cdot \frac{n_{i+}}{n} \cdot \frac{n_{+j}}{n} = \frac{n_{i+} \cdot n_{+j}}{n}$,

$$\hat{P}_{ij} = \frac{n_{ij}}{n}, \hat{P}_{i+} = \sum_{j=1}^J \frac{n_{ij}}{n} = \frac{n_{i+}}{n}$$

$\therefore (n_{11}, n_{12}, n_{21}, n_{22}) \sim \text{Mul}(n, P_{11}, P_{12}, P_{21}, P_{22})$

(3) Reject H_0 if $X^2 > \chi^2_{(I-1)(J-1), \alpha}$

If $Y \sim \chi^2_\gamma$, then $\frac{Y - \gamma}{\sqrt{2\gamma}} \rightarrow N(0, 1)$ as $\gamma \rightarrow \infty$

Two-way Contingency Tables

*df = the number of parameters – the number of parameters under H_0

$(n_{11}, n_{12}, n_{21}, n_{22}) \sim \text{Mul}(n, P_{11}, P_{12}, P_{21}, P_{22})$ for 2×2 table; $\sum_{i=1}^I \sum_{j=1}^J P_{ij} = 1$

(1) $\Omega = \text{parameter space} = \{P_{ij} \mid \sum_{i=1}^I \sum_{j=1}^J P_{ij} = 1\}$

$\Rightarrow$ the number of parameters in $\Omega = I \times J - 1$

(2) $\Theta_0 = \text{parameter space under } H_0 = \{P_{ij} \mid P_{ij} = P_{i+} \cdot P_{+j}\}$

$\Rightarrow$ the number of parameters in $\Omega = (I - 1) + (J - 1)$ ($\because \sum_{i=1}^I P_{i+} = 1, \sum_{j=1}^J P_{+j} = 1$)

$\therefore \text{df} = (IJ - 1) - [(I - 1) + (J - 1)] = (I - 1)(J - 1)$

Two-way Contingency Tables

*The likelihood-ratio Chi-square statistic :

$$(1) H_0 : P_{ij} = P_{i+} \cdot P_{+j}$$

$$(2) G^2 = -2\log(\Lambda) = 2 \sum_{i=1}^I \sum_{j=1}^J \frac{n_{ij}}{\hat{\mu}_{ij}} \sim \chi^2_{(I-1)(J-1)} \text{ under } H_0 \text{ if } n_{ij} \text{ are large,}$$

$$\text{where } \Lambda = \frac{\text{maximum likelihood when parameters satisfy } H_0}{\text{maximum likelihood when parameters are unrestricted}} = \frac{\max_{H_0} L(P_{ij})}{\max_{\Omega} L(P_{ij})},$$

$$L(P_{ij}) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J P_{ij}^{n_{ij}}$$

$$(3) \text{ Reject } H_0 \text{ if } G^2 > \chi^2_{(I-1)(J-1), \alpha} = 3.84, \alpha = 0.05$$

$$\hat{\mu}_{ij} = n \cdot \hat{P}_{ij} = \frac{n_{i+} \cdot n_{+j}}{n}$$

Two-way Contingency Tables

e. g.

		Party Identification		
		Democrat	Independent	Republican
Sex	F	279 (261.4)	73 (70.7)	255 (244.9)
	M	165 (182.3)	47 (49.3)	191 (171.1)
		444	120	416
				980

$$\hat{\mu}_{ij} = n \cdot \hat{P}_{ij} = \frac{n_{i+} \cdot n_{+j}}{n} \text{ (e. g. } \hat{\mu}_{11} = \frac{577 \times 444}{980} = 261.4 \text{)}$$

$$\chi^2 = 7.01 > \chi^2_{2,0.05} = 3.84$$

$$G^2 = 7.00 > \chi^2_{2,0.05} = 3.84, p\text{-value} = P(\chi^2 > 7.0) = 0.028 \text{ (準確度至小數點下不為零後的2位數即可)}$$

$\therefore$ Reject $H_0 \Rightarrow$ 證據足以顯示性別與認同的政黨有關

Two-way Contingency Tables

*adjusted residuals = $e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\sqrt{\hat{\mu}_{ij}(1 - \hat{p}_{i+})(1 - \hat{p}_{+j})}} \sim N(0, 1)$ under $H_0 : \theta = 1$

$\Rightarrow$ lack of fit H_0 if $|\text{adjusted residuals}| > 2$ or 3

e. g.

		Party Identification		
		Democrat	Independen t	Republican
Sex	F	279 (2.29)	73 (0.46)	255 (-2.62)
	M	165 (-2.29)	47 (-0.46)	191 (2.62)

$\because \sum_{i=1}^2 n_{ij} = \sum_{i=1}^2 \hat{\mu}_{ij}$ and $\sum_{j=1}^3 n_{ij} = \sum_{j=1}^3 \hat{\mu}_{ij}$
 $\therefore \sum_{i=1}^2 (n_{ij} - \hat{\mu}_{ij}) = 0$ and $\sum_{j=1}^3 (n_{ij} - \hat{\mu}_{ij}) = 0$
 (即 $n_{1j} - \hat{\mu}_{1j}$ 和 $n_{2j} - \hat{\mu}_{2j}$ 有相同值但符號相反)

$$\hat{P}_{1+} = \frac{577}{980} \approx 0.589, \hat{P}_{+1} = \frac{444}{980} \approx 0.453 \Rightarrow e_{11} = \frac{279 \times 261.4a}{\sqrt{261.4(1-0.589)(1-0.453)}} = 2.29$$

	Democrat	Independent
F	279	73
M	165	47

$$\hat{\theta} = \frac{279 \times 191}{165 \times 225} \approx 1.44$$

$\Rightarrow$ 女性認同共和黨對民主黨的勝算較男性的勝算高出 44%

Two-way Contingency Tables

*Partitioning Chi-squared :

	Democrat	Independent
F	279	73
M	165	47

$$G_1^2 = 0.16$$

		D+I	Republican
F		279+73	225
M		165+47	191

$$G_2^2 = 6.84$$

$$G^2 = G_1^2 + G_2^2 = 0.16 + 6.84 = 7.0 \text{ (why?)}$$

	Democrat	Republican
F	279	225
M	165	191

$$* G_1^2 = ?$$

$$G^2 = * G_1^2 + * G_2^2 ?$$

	Independent	Republican
F	73	225
M	47	191

$$* G_2^2 = ?$$

$$X^2 = X_1^2 + X_2^2 ? \text{ (No!)}$$

Two-way Contingency Tables

	D	I	R	
F	n_{11}	n_{12}	n_{13}	n_{1+}
M	n_{21}	n_{22}	n_{23}	n_{2+}
	n_{+1}	n_{+2}	n_{+3}	

$$G^2 = \sum_{i=1}^2 \sum_{j=1}^3 n_{ij} \cdot \log \frac{n_{ij}}{\hat{\mu}_{ij}}, \quad \hat{\mu}_{ij} = \frac{n_{i+} \cdot n_{+j}}{n}$$

$$G_1^2 = n_{11} \log \frac{n_{11}}{\hat{\mu}_{11}^{(1)}} + n_{12} \log \frac{n_{12}}{\hat{\mu}_{12}^{(1)}} + n_{21} \log \frac{n_{21}}{\hat{\mu}_{21}^{(1)}} + n_{22} \log \frac{n_{22}}{\hat{\mu}_{22}^{(1)}}$$

$$\hat{\mu}_{ij}^{(1)} = \frac{(n_{i1} + n_{i2}) \cdot n_{+j}}{n_{11} + n_{12} + n_{21} + n_{22}}$$

$$G_2^2 = (n_{11} + n_{12}) \log \frac{(n_{11} + n_{12})}{\hat{\mu}_{11}^{(2)}} + (n_{21} + n_{22}) \log \frac{(n_{21} + n_{22})}{\hat{\mu}_{21}^{(2)}} + n_{13} \log \frac{n_{13}}{\hat{\mu}_{12}^{(2)}} + n_{23} \log \frac{n_{23}}{\hat{\mu}_{22}^{(2)}}$$

$$\hat{\mu}_{i1}^{(2)} = \frac{n_{i+} \cdot (n_{11} + n_{12} + n_{21} + n_{22})}{n}, \quad \hat{\mu}_{i2}^{(2)} = \frac{n_{i+} \cdot n_{+3}}{n}$$

· Note : The overall Pearson X^2 statistic does not equal the sum of X^2 values for the separate tables in a partition.

Two-way Contingency Tables

*若 n 很大時， X^2 收斂較 G^2 快。

*當 $\frac{n}{I \cdot J} < 5$ ， G^2 近似得較差。

*若 I, J 很大使得 $\hat{\mu}_{ij} \approx 1$ ， X^2 近似卡方分配效果很差。

Two-way Contingency Tables

*Cochran-Mantel-Haenszel test : (適用於 Ordinal-Ordinal data)

(1) H_0 : independence

H_1 : nonzero correlation (linear trend)

(2) $M^2 = (n - 1) \cdot r^2 \approx \chi_1^2$ for $I \times J$ table, large n ,

$$\text{where } r = \hat{\rho} = \frac{\sum_i \sum_j u_i v_j n_{ij} - (\sum_i u_i n_{i+})(\sum_j v_j n_{+j})/n}{\sqrt{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n} \right] \left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n} \right]}}$$

$u_1 \leq u_2 \leq \dots \leq u_I$: scores for the rows,

$v_1 \leq v_2 \leq \dots \leq v_J$: scores for the columns.

· 一般 Score 的選取對 M^2 影響不大。

但若 cell count 有的很小，有的很大，不同的 Score， M^2 值相差很大。

Two-way Contingency Tables

*Cochran-Mantel-Haenszel test : (For Ordinal-Ordinal data)

$$\cdot M = \sqrt{n-1} \cdot r \sim N(0, 1)$$

$$H_0 : \rho = 0 \text{ vs. } H_1 : \rho > 0 \Rightarrow \text{Reject } H_0 \text{ if } M > Z_\alpha$$

· 行變數和列變數交換， M^2 值不變。(i. e. 對 $I \times J$ table $\Rightarrow J \times I$ table， M^2 值不變)

· Note : CMH test 的 $df = 1$ 而非 $(I-1)(J-1)$ $\because H_1$ 為線性相關而非一般的相關

· 當資料中 X 和 Y 有線性相關，使用 M^2 的檢定力較 χ^2 和 G^2 大。

· Score 的選取通常依經驗判斷，equally-spaced scores 通常較合理。

$$\cdot \text{Spearman's } \rho \tilde{\rho} = \frac{\sum (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum (R_i - \bar{R})^2 \sum (S_i - \bar{S})^2}}, \text{ where } R_i : \text{rank of } X_i, S_i : \text{rank of } Y_i$$

Two-way Contingency Tables

*linear correlation coefficient = $\rho = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

sample correlation coefficient = $\hat{\rho} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}} = \frac{S_{xy}}{S_x S_y}$

*把類別變數視為連續變數：

		Y			
		v_1	v_2	...	v_J
		1	2	...	J
X	u_1	n_{11}	n_{12}	...	n_{1J}
	u_2	n_{21}	n_{22}	...	n_{2J}
	...	...	...	...	...
	u_I	n_{I1}	n_{I2}	...	n_{IJ}
	...	...	...	...	...



i	X	Y
1	u_1	v_1
...	...	...
$n_{11}+1$	u_1	v_1
$n_{11}+1$	u_2	v_1
...	...	...
$n_{11}+n_{21}$	u_2	v_1
...	...	...

* $-1 \leq \rho \leq 1$

* $\rho = 0$ 代表 X 和 Y 無線性關係 (注意：不是無關)

$\rho = 1$ 代表 X 和 Y 線性正相關

(i.e. $Y = ax + b$, $a > 0$)

$\rho = -1$ 代表 X 和 Y 線性負相關

(i.e. $Y = ax + b$, $a < 0$)

Two-way Contingency Tables

e. g. 母親每日平均飲酒量和嬰兒先天性畸形是否有關?

		Malformation		
		Absent	Present	
Alcohol Consumption	0	17066	48	17114
	<1	14464	38	14502
	1-2	788	5	793
	3-5	126	1	127
	≥6	37	1	38



Percentage (%)	adjusted residual
0.28	-0.18
0.26	-0.71
0.63	1.84
0.79	1.06
2.63	2.71

$\chi^2 = 12.1$, p-value = 0.02 < $\alpha = 0.05$

$G^2 = 6.2$, p-value = 0.19 > $\alpha = 0.05$

$M^2 = 6.6$, p-value = 0.01 < $\alpha = 0.05$ (or $M = 2.56$, p-value = 0.005 < $\alpha = 0.05$)

($v_1 = 0, v_2 = 0.5, v_3 = 1.5, v_4 = 4.0, v_5 = 7$, midpoints of categories)

$M^2 = 1.83$, p-value = 0.18 ($v_1 = 1, v_2 = 2, v_3 = 3, v_4 = 4, v_5 = 5$)

Two-way Contingency Tables

*Trend tests for $2 \times J$ tables

(1) $u_1 = 0, u_2 = 1$: scores for X

$v_1 \leq v_2 \leq \dots \leq v_J$: scores for Y

$\therefore \sum u_i v_i n_{ij} = \sum v_j n_{2j}$: total scores on Y in row 2

$$\therefore r = \frac{\sum v_j n_{2j} - n_{2+}(\sum v_j n_{+j})/n}{\sqrt{\left[n_{2+} - \frac{(n_{2+})^2}{n}\right] \left[\sum v_j^2 n_{+j} - \frac{(\sum v_j n_{+j})^2}{n}\right]}}$$

$$= \frac{\sum v_j n_{2j} / n_{2+} - (\sum v_j n_{+j})/n}{\sqrt{\left[n_{2+} - \frac{(n_{2+})^2}{n}\right] \left[\sum v_j^2 n_{+j} - \frac{(\sum v_j n_{+j})^2}{n}\right] / n_{2+}}}$$

		Y				
		v_1	v_2	...	v_J	
		1	2	...	J	
X	1	n_{11}	n_{12}	...	n_{1J}	n_{1+}
	2	n_{21}	n_{22}	...	n_{2J}	n_{2+}
		n_{+1}	n_{+2}	...	n_{+J}	

difference between two row means of scores of Y

Two-way Contingency Tables

*Trend tests for $2 \times J$ tables

(2) Mann Whitney test (or Wilcoxon Rank-Sum test) :

H_0 : 兩組樣本母體的分配相同

H_1 : 兩組樣本母體的分配不同

· 檢定統計量 : $T = S - \frac{(n_1+1)n_1}{2}$, 其中S 為將兩組樣本混合排序 , 加總其中一組的 rank

· 若 $n \leq 20$, reject H_0 if $T < W_{\frac{\alpha}{2}}$ or $T > W_{1-\frac{\alpha}{2}}$ (critical value W_{α} 須查表)

若 $n > 20$, 令 $Z = \frac{T-E(T)}{\sigma_T}$, $E(T) = \frac{n_1(n+1)}{2}$, $\sigma_T^2 = \frac{n_1 n_2 (n+1)}{12}$, reject H_0 if $|Z| > Z_{\frac{\alpha}{2}}$

· 當 $v_1, v_2, \dots, v_J$ 為 midranks , $M^2 \equiv$ detect mean ranks for the two rows.

$\Rightarrow Z^2$ is equivalent to M^2

· The ordinal test M^2 for $I \times 2$ tables is called the Cochran-Armitage trend test.

當兩母體分配 Shape 相同、變異相同 ,
假設變成檢定中位數 :

$H_0 : M_X = M_Y$ vs. $H_1 : M_X \neq M_Y$

Two-way Contingency Tables

* M^2 只適用於 Ordinal-Ordinal 列連表或 Nominal-Ordinal (Ordinal-Nominal) 列連表，

但 Nominal 變數的 level 只能有 2 類。當 Nominal 變數的 level 有 3 類以上時，可使用：

(1) CMH test

$H_0 : X \text{ is independent of } Y \equiv H_0 : \mu_1 = \mu_2 = \dots = \mu_I$

$$\text{test statistic} = \text{CMH} = (n-1) \frac{\sum_j n_{i+} (\bar{y}_i - \bar{y})^2}{\sum_j v_j^2 n_{+j} - n \bar{y}^2} \equiv \frac{\text{SSB}}{\text{SSW}},$$

where $\bar{y}_i = \sum_j v_j n_{ij} / n_{i+}$: the sample mean of Y in row i

$$\bar{y} = \sum_j v_j n_{+j} / n : \text{the overall mean}$$

$$v_1 \leq v_2 \leq \dots \leq v_J : \text{scores for Y}$$

若 $Y_{ij} \sim N(\mu_i, \sigma^2), i = 1, \dots, I$ ，

此檢定類似 One-way ANOVA

		Y				
		v_1	v_2	...	v_J	
		1	2	...	J	Score mean
X	1	n_{11}	n_{12}	...	n_{1J}	$\bar{y}_1$
	2	n_{21}	n_{22}	...	n_{2J}	$\bar{y}_2$
	...	...	...	...	...	...
	I	n_{I1}	n_{I2}	...	n_{IJ}	$\bar{y}_I$
		n_{+1}	n_{+2}	...	n_{+J}	$\bar{y}$

Two-way Contingency Tables

* M^2 只適用於 Ordinal-Ordinal 列連表或 Nominal-Ordinal (Ordinal-Nominal) 列連表，

但 Nominal 變數的 level 只能有 2 類。當 Nominal 變數的 level 有 3 類以上時，可使用：

(2) Kruskal-Wallis test

$H_0 : I \text{ 組資料都是來自相同分配}$

$$\text{test statistic} = X^2 = \frac{12}{N(N+1)} \sum_{i=1}^I \frac{1}{n_i} \left[R_i - \frac{n_i(n+1)}{2} \right]^2 \approx X_{I-1}^2,$$

R_i 為全體樣本混合排序，第 i 組樣本的 rank 和

*當 $v_1, v_2, \dots, v_J$ 為 midranks，CMH 檢定相當於 Kruskal-Wallis 檢定（比較 I 組的 mean rank 是否相等）

*當 $I = 2$ ， $\text{CMH} = M^2$

Two-way Contingency Tables

- 針對任一種檢定，報告 p-value 比只報告 reject H_0 或 do not reject H_0 好。
- 對於檢定統計量近似卡方分配，當小樣本時，現有統計軟體 statXact 可生成檢定統計量的 exact sampling distribution，並計算 exact p-value

e. g.

		Y								
X	1	7	0	0	0	0	0	0	1	1
	2	1	1	1	1	1	1	0	0	
	3	8	0	0	0	0	0	0	0	0

稀疏資料 (Sparse Data)

3×9 table

$$*H_0 : P_{ij} = P_{i+} \cdot P_{+j} \text{ vs. } H_1 : P_{ij} \neq P_{i+} \cdot P_{+j}$$

$$*X^2 = 22.3, df = 16$$

$$*p - \text{value} = 0.13 > 0.05$$

$$* \text{exact } p - \text{value} = 0.001 < 0.05$$

Two-way Contingency Tables

*Fisher's Exact test (for 2×2 tables)

		Y		
		1	2	
X	1	n_{11}	n_{12}	n_{1+}
	2	n_{21}	n_{22}	n_{2+}
		n_{+1}	n_{+2}	

$$(1) H_0 : \theta = 1 \text{ vs. } H_1 : \theta > 1 \text{ [i.e. } P_1 > P_2, P_i = P(Y = 1|X = i)]$$

$$(2) \text{ test statistic : } (N_{11}|n_{1+}, n_{2+}, n_{+1}, n_{+2}) \sim \text{Hypergeometric under } H_0 : \theta = 1$$

$$\text{i.e. } P(N_{11} = k) = \frac{\binom{n_{1+}}{k} \binom{n_{2+}}{n_{+1}-k}}{\binom{n}{n_{+1}}}, \text{ where } \binom{a}{b} = \frac{a!}{b!(a-b)!}$$

$$(3) \text{ Reject } H_0 \text{ if } n_{11} \geq c, c \text{ is chosen by } \alpha \geq P(N_{11} \geq c|\theta = 1) \Rightarrow p - \text{value} = P(N_{11} \geq n_{11})$$

*A $100(1 - \alpha)\%$ C.I. for θ is $\{\theta | P(N_{11} \geq n_{11}) > \alpha\}$

但實際上的 Confidence level 可能會超過既定的 $100(1 - \alpha)\%$
可改用 mid p-value 計算

Two-way Contingency Tables

*Fisher’s Exact test (for 2 × 2 tables)

e. g. Fisher’s Tea-Tasting Experiment

Fisher 的同事宣稱只要喝一口奶茶即可知道是先加牛奶或先加紅茶。

		Guess		
		Milk	Tea	
Poured First	Milk	3	1	4
	Tea	1	3	4
		4	4	8

Fisher 設計一實驗測試：有 8 杯奶茶，已知 4 杯是先加牛奶，4杯是先加紅茶。

今將 8 杯奶茶隨機排列讓 Fisher 的同事品嚐並猜測，試問是否有證據顯示猜測和奶茶調製順序有關？

* $H_0 : \theta = 1$ vs. $H_1 : \theta > 1$

*p – value = $P(N_{11} \geq 3) = P(N_{11} = 3) + P(N_{11} = 4)$
 $= \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} + \frac{\binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = 0.229 + 0.014 = 0.243 \geq \alpha = 0.05$
 $\Rightarrow$ Do not reject H_0

Two-way Contingency Tables

0	4	1	3	2	2	3	1	4	0
4	0	3	1	2	2	1	3	0	4

n_{i1}	probability	p-value for $\begin{cases} H_0 : \theta = 1 \\ H_1 : \theta > 1 \end{cases}$	$X^2 = \sum_{i=1}^2 \sum_{j=1}^2 \frac{(n_{ij}-2)^2}{2}$	$\hat{\mu}_{ij} = \frac{4 \times 4}{8} = 2$
0	0.014	1.0	8.0	
1	0.229	0.986	2.0	
2	0.514	0.757	0.0	
3	0.229	0.243	2.0	
4	0.014	0.014	8.0	

*If $\alpha = 0.05$, critical value $c = 4 \Rightarrow P(\text{Type I error}) = 0.014$ (太保守 !)

An exact 95% C.I for θ is (0.21, 626.17) $\rightarrow$ (0.31, 308.55) (改用 mid p-value)

Two-way Contingency Tables

*Fisher's Exact test (for 2×2 tables)

· P-values for the two-sided test of test statistic N_{11}

$$H_0 : \theta = 1 \text{ vs. } H_1 : \theta \neq 1$$

$$p\text{-value} = \sum_y P(N_{11} = y), \text{ where } P(N_{11} = y) \leq P(N_{11} = n_{11}) \quad n_{11} : \text{observed value}$$

e. g. Fisher's Tea-Tasting data for testing $\begin{cases} H_0 : \theta = 1 \\ H_1 : \theta \neq 1 \end{cases}$

$$\begin{aligned} p\text{-value} &= \sum_y P(N_{11} = y), \text{ where } P(N_{11} = y) \leq P(N_{11} = 3) = 0.229 \\ &= P(N_{11} = 0) + P(N_{11} = 1) + P(N_{11} = 3) + P(N_{11} = 4) = 0.486 \end{aligned}$$

Two-way Contingency Tables

*Fisher's Exact test (for 2×2 tables)

· 當列的和或行的和相等 (i. e. $n_{1+} = n_{2+}$ or $n_{+1} = n_{+2}$) 雙尾檢定的 P 值是單尾檢定 P 值的 2 倍。

· 由於檢定統計量 $N_{11} \sim \text{hypergeometric}$ 為 Discrete distribution，故雖然型 I 誤差 $\alpha = 0.05$ ，

但實際上型 I 誤差常會不到 0.05，因此可以說 Fisher's exact test is conservative. (太保守的檢定)

Two-way Contingency Tables

*Fisher's Exact test (for 2×2 tables)

· P-values for the two-sided test of test statistic χ^2

$$H_0 : \theta = 1 \text{ vs. } H_1 : \theta \neq 1$$

$$p\text{-value} = \sum_y P(X^2 = y), \text{ where } P(X^2 = y) \geq P(X^2 = \chi^2) \quad \chi^2 : \text{observed value}$$

e. g. Fisher's Tea-Tasting data for testing $\begin{cases} H_0 : \theta = 1 \\ H_1 : \theta \neq 1 \end{cases}$

$$p\text{-value} = \sum_y P(X^2 \geq 2.0) = 0.014 + 0.014 + 0.229 + 0.229 = 0.486$$

Two-way Contingency Tables

*當檢定統計量的分配為離散時，其 P 值的平均數通常很大。

e. g. Fisher's Tea-Tasting data for testing $\begin{cases} H_0 : \theta = 1 \\ H_1 : \theta > 1 \end{cases}$

The expected value of the p-value

$$= 0.014 \times 0.014 + 0.243 \times 0.229 + 0.757 \times 0.514 + 0.986 \times 0.229 + 1.0 \times 0.014$$

$$= 0.685 > 0.5$$

Two-way Contingency Tables

*The mid p-value equals half the probability of the observed result plus the probability of more extreme results.

$$\text{e. g. } \begin{cases} H_0 : \theta = 1 \\ H_1 : \theta > 1 \end{cases} \quad \text{mid p-value} = \frac{1}{2}P(N_{11} = 3) + P(N_{11} = 4) = \frac{1}{2} \times 0.229 + 0.014 = 0.129$$

$$\begin{cases} H_0 : \theta = 1 \\ H_1 : \theta \neq 1 \end{cases} \quad \text{mid p-value} = \frac{1}{2}P(X^2 = 2) + P(X^2 = 8) = 0.229 + 2 \times 0.014 = 0.257$$

· Mid p-value 的缺點：不保證型I誤差小於既定的 α 值；優點：較不保守。